Global Literacy & Education Trends
Day 1: Project Setup & Data Collection
Environment Setup (Google Colab)
Download 5 CSV datasets from OWID
Adult Literacy Rate
Youth Literacy Rate (Male, Female)
Illiterate Population (Total, Male, Female)
GDP per Capita
Average Years of Schooling
Initial data loading into DataFrames
Day 2: Data Understanding & Exploration
Load datasets into separate DataFrames
Understand data structure and shape
Check data types and basic statistics
Identify common columns for merging
Document initial observations
Day 3: Data Cleaning
Handle missing values (drop/impute)
Remove duplicates
Standardize country names
Identify unusual values
Rename columns for clarity
Filter years (1990-2023)
Day 4: Feature Engineering
Create Illiteracy %
Calculate Literacy Gender Gap
Compute GDP per Schooling Year
Build Education Index
Calculate Youth Literacy Average
Measure Literacy Growth Rate
Additional custom features
Day 5: Exploratory Data Analysis (EDA)
Univariate Analysis (histograms, box plots, bar charts)
Bivariate Analysis (scatter plots, heatmaps, correlations)
Time series trends analysis
Identify outliers and patterns
Create visualizations with Matplotlib/Seaborn/Plotly
Document insights

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Day 6: SQL Database Design & Queries
Create 3 SQL tables (literacy_rates, illiteracy_population, gdp_schooling)
Define composite keys (country, year)
Insert cleaned data
Write 13 SQL queries
Single table queries (1-9)
Join queries (10-13)
Test and validate query outputs
Day 7: Dashboard Development (Power BI or Streamlit)
Power BI Option
Import SQL query outputs
Create custom visuals
Add slicers for year/country
Build comprehensive dashboard
Streamlit Option
SQL Query Executor page
EDA Visualizations page
Country Profile page
Day 8: Visualization & Insights
Create 10 key visualizations
Literate vs Illiterate trends
Adult vs Youth literacy gap
Gender disparities
Literacy vs GDP correlation
Schooling years vs literacy
Top/bottom performers ranking
Continental patterns
Illiteracy and population size
Long-term projections
Impact of crises on literacy
Day 9: Final Review & Submission
Review all deliverables
Prepare 3 cleaned DataFrames
Finalize SQL scripts
Complete Jupyter Notebook with EDA
Finalize Power BI Dashboard or Streamlit App
Write project summary with findings
Final quality check and submission